SHETH LUJ AND SIR MV COLLEGE
Subject: Data Analysis with SAS / SPSS /R

Practical No: 11

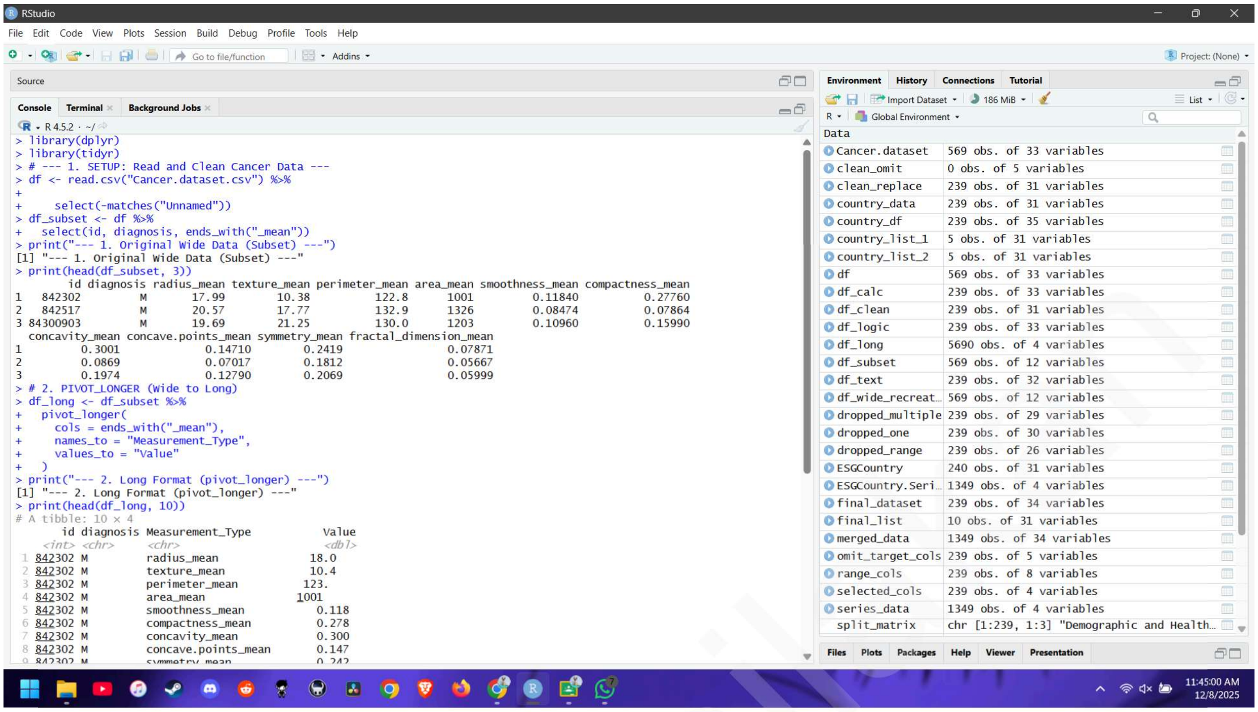
Aim: Reshaping data using pivot_longer()/pivot_wider() (R).

Code:

```
library(dplyr)
library(tidyr)

# --- 1. SETUP: Read and Clean Cancer Data ---
df <- read.csv("Cancer.dataset.csv") %>%

    select(-matches("Unnamed"))

df_subset <- df %>%
  select(id, diagnosis, ends_with("_mean"))

print("--- 1. Original Wide Data (Subset) ---")
print(head(df_subset, 3))

# 2. PIVOT_LONGER (Wide to Long)
df_long <- df_subset %>%
  pivot_longer(
    cols = ends_with("_mean"),
    names_to = "Measurement_Type",
    values_to = "Value"
  )

print("--- 2. Long Format (pivot_longer) ---")
print(head(df_long, 10))

# 3. PIVOT_WIDER (Long to Wide)
df_wide_recreated <- df_long %>%
  pivot_wider(
    names_from = Measurement_Type,
    values_from = Value
  )

print("--- 3. Wide Format (Back to Original) ---")
print(head(df_wide_recreated, 3))
```

Omith Thilakan
S097

Output:


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Console Terminal Background Jobs
R 4.5.2 · ~/
> library(dplyr)
> library(tidyr)
> # --- 1. SETUP: Read and Clean Cancer Data ---
> df <- read.csv("Cancer.dataset.csv") %>%
+ select(-matches("Unnamed"))
> df_subset <- df %>%
+ select(id, diagnosis, ends_with("_mean"))
> print("--- 1. Original Wide Data (Subset) ---")
[1] "--- 1. Original Wide Data (Subset) ---"
> print(head(df_subset, 3))
id diagnosis radius_mean texture_mean perimeter_mean area_mean smoothness_mean compactness_mean
1 842302 M 17.99 10.38 122.8 1001 0.11840 0.27760
2 842517 M 20.57 17.77 132.9 1326 0.08474 0.07864
3 84300903 M 19.69 21.25 130.0 1203 0.10960 0.15990
concavity_mean concave.points_mean symmetry_mean fractal_dimension_mean
1 0.3001 0.14710 0.2419 0.07871
2 0.0869 0.07017 0.1812 0.05667
3 0.1974 0.12790 0.2069 0.05999
> # 2. PIVOT_LONGER (Wide to Long)
> df_long <- df_subset %>%
+ pivot_longer(
+ cols = ends_with("_mean"),
+ names_to = "Measurement_Type",
+ values_to = "Value"
+)
> print("--- 2. Long Format (pivot_longer) ---")
[1] "--- 2. Long Format (pivot_longer) ---"
> print(head(df_long, 10))
A tibble: 10 × 4
id diagnosis Measurement_Type Value
<int> <chr> <chr> <dbl>
1 842302 M radius_mean 18.0
2 842302 M texture_mean 10.4
3 842302 M perimeter_mean 123.
4 842302 M area_mean 1001
5 842302 M smoothness_mean 0.118
6 842302 M compactness_mean 0.278
7 842302 M concavity_mean 0.300
8 842302 M concave.points_mean 0.147
Environment History Connections Tutorial
Import Dataset 186 MiB List
Global Environment
Data
Cancer.dataset 569 obs. of 33 variables
clean_omit 0 obs. of 5 variables
clean_replace 239 obs. of 31 variables
country_data 239 obs. of 31 variables
country_df 239 obs. of 35 variables
country_list_1 5 obs. of 31 variables
country_list_2 5 obs. of 31 variables
df 569 obs. of 33 variables
df_calc 239 obs. of 33 variables
df_clean 239 obs. of 31 variables
df_logic 239 obs. of 33 variables
df_long 5690 obs. of 4 variables
df_subset 569 obs. of 12 variables
df_text 239 obs. of 32 variables
df_wide_recreat… 569 obs. of 12 variables
dropped_multiple 239 obs. of 29 variables
dropped_one 239 obs. of 30 variables
dropped_range 239 obs. of 26 variables
ESGCountry 240 obs. of 31 variables
ESGCountry.Seri… 1349 obs. of 4 variables
final_dataset 239 obs. of 34 variables
final_list 10 obs. of 31 variables
merged_data 1349 obs. of 34 variables
omit_target_cols 239 obs. of 5 variables
range_cols 239 obs. of 8 variables
selected_cols 239 obs. of 4 variables
series_data 1349 obs. of 4 variables
split_matrix chr [1:239, 1:3] "Demographic and Health…
Files Plots Packages Help Viewer Presentation
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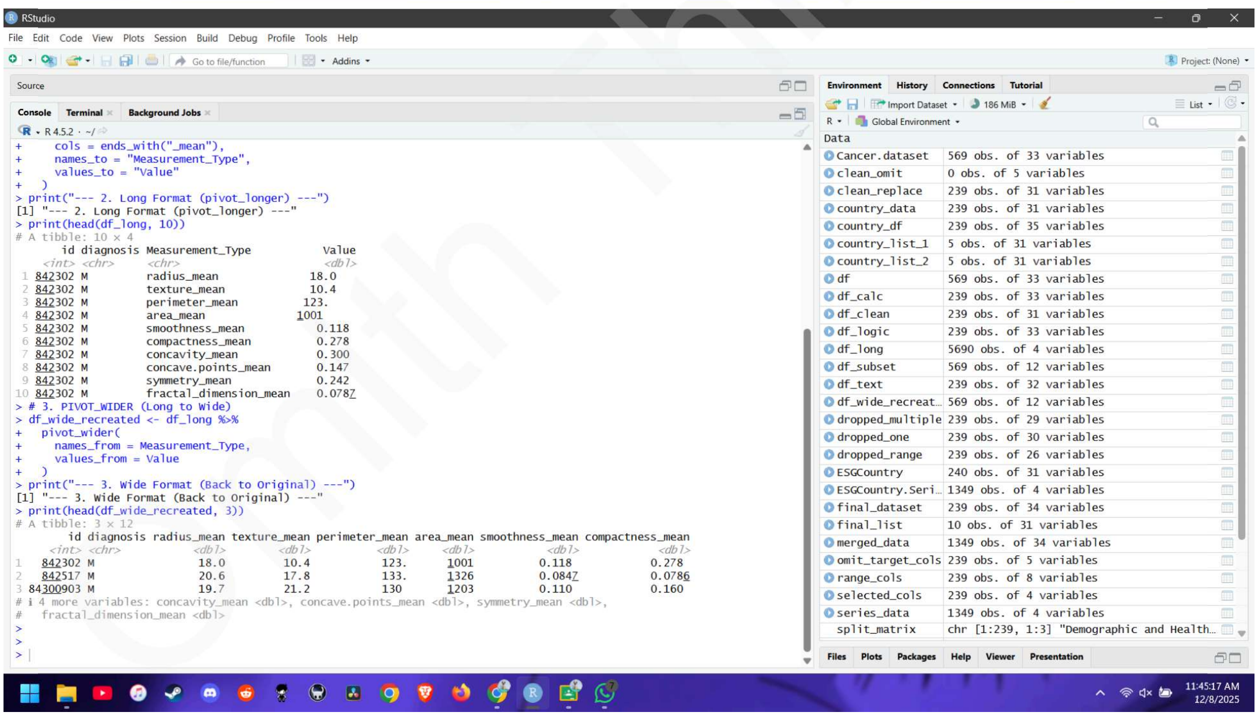
RStudio
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Console Terminal Background Jobs
R 4.5.2 · ~/
+ cols = ends_with("_mean"),
+ names_to = "Measurement_Type",
+ values_to = "Value"
+)
> print("--- 2. Long Format (pivot_longer) ---")
[1] "--- 2. Long Format (pivot_longer) ---"
> print(head(df_long, 10))
A tibble: 10 × 4
id diagnosis Measurement_Type Value
<int> <chr> <chr> <dbl>
1 842302 M radius_mean 18.0
2 842302 M texture_mean 10.4
3 842302 M perimeter_mean 123.
4 842302 M area_mean 1001
5 842302 M smoothness_mean 0.118
6 842302 M compactness_mean 0.278
7 842302 M concavity_mean 0.300
8 842302 M concave.points_mean 0.147
9 842302 M symmetry_mean 0.242
10 842302 M fractal_dimension_mean 0.0787
> # 3. PIVOT_WIDER (Long to Wide)
> df_wide_recreated <- df_long %>%
+ pivot_wider(
+ names_from = Measurement_Type,
+ values_from = Value
+)
> print("--- 3. Wide Format (Back to Original) ---")
[1] "--- 3. Wide Format (Back to Original) ---"
> print(head(df_wide_recreated, 3))
A tibble: 3 × 12
id diagnosis radius_mean texture_mean perimeter_mean area_mean smoothness_mean compactness_mean
<int> <chr> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1 842302 M 18.0 10.4 123. 1001 0.118 0.278
2 842517 M 20.6 17.8 133. 1326 0.0847 0.0786
3 84300903 M 19.7 21.2 130 1203 0.110 0.160
i 4 more variables: concavity_mean <dbl>, concave.points_mean <dbl>, symmetry_mean <dbl>,
fractal_dimension_mean <dbl>
Environment History Connections Tutorial
Global Environment
Data
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clean_omit 0 obs. of 5 variables
clean_replace 239 obs. of 31 variables
country_data 239 obs. of 31 variables
country_df 239 obs. of 35 variables
country_list_1 5 obs. of 31 variables
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ESGCountry.Seri… 1349 obs. of 4 variables
final_dataset 239 obs. of 34 variables
final_list 10 obs. of 31 variables
merged_data 1349 obs. of 34 variables
omit_target_cols 239 obs. of 5 variables
range_cols 239 obs. of 8 variables
selected_cols 239 obs. of 4 variables
series_data 1349 obs. of 4 variables
split_matrix chr [1:239, 1:3] "Demographic and Health…
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